Engineering-proof coverage audit

Scope and mechanism

The runnable audit is **Engineering Lab → Model validation → Engineering-proof coverage audit**. src/simulation/proofCoverage.ts inventories the implemented consequence families and canonical runtime numeric fields. It checks equation/provenance/units/model version/quality/sensitivity/limitations/validation/trace completeness, and flags newly introduced unmapped runtime numbers. Tests exercise initial and disrupted worlds and deliberately introduce an unmapped output to verify detection.

The inventory covers pump, sump, feeder, zone, sensor and asset runtime numbers; hydraulic results; water and material ledgers; material throughput; site aggregates; and the five context-bound analysis families below. Clock/sample timestamps and UI counts are metadata. This is a scoped audit of the implemented engineering increment, not a claim that static inspection formally proves every frontend expression or future code path correct.

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Consequence-to-proof matrix

Predicted/displayed engineering consequence	Actual implementation	Proof target / UI access
Pump flow, head, friction, efficiency, speed limiting and power	hydraulics.ts, plant.ts	Pump asset ID; analyser, inspector, schematic or audit
Integrated pumping energy and tariff cost	engine.ts pump counters	Pump asset ID and site:energy; Energy or audit
Stored water, stage, overflow and mass residual	water.ts, waterNetwork.ts	Sump asset ID and site:water; DeepWater, resilience or audit
Inter-sump transfers and cumulative mass closure	Recorded water ledger/totals	Sump asset ID / site:water; transfer debits/credits shown once
Source/feeder capacity, motor reservations and unsupplied demand	plant.ts	Feeder asset ID / site:power; Energy, inspector or audit
Delayed zone restrictions, restart ramp, foregone hours/tonnes/value	production.ts, engine.ts	Zone asset ID; Production, inspector, schematic or audit
Primary/alternate route flow, common hoist and finite receiving inventory	materialHandling.ts	Route ID / mat-hoist; material controls, schematic or audit
Processing throughput, actual processed/deferred tonnes and inventory	Material ledger / plant state	Plant asset ID / site:production; Production or audit
Reported readings, missing values, residuals and quality	sensors.ts	Sensor asset ID / site:quality; inspector or audit
Generic conditional status and modeled functional fractions	propagation.ts and physical model results	site:dependencies; audit. Non-physical fractions remain declared approximations.
Forecast envelopes and threshold arrivals	forecast.ts, canonical projection	analysis:forecast; Forecast and DeepWater, bound to displayed forecast points
Recovery onset/confirmation, exposure, cost and revenue consequences	engineeringStudies.ts, recovery.ts	analysis:recovery; each comparison/sensitivity result
Inspection residuals, hypothesis dispositions and restored-capacity comparisons	diagnostics.ts	analysis:diagnostics; workbench/challenges, bound to the actual report
Evidence/claims/calculation/alternative/action scoring	investigation.ts	analysis:challenge; reviewed challenge rubric
Capital assumptions, operating expenses and avoided deferred tonnes	Matched investment projections	analysis:investment; each investment row

Important distinctions

• Snapshot proofs use recorded scenario inputs and effective assumptions. Context-bound study proofs consume the actual computed result, not a generic asset explanation.

• Recovery onset is measured from projected performance. Electricity restoration or a short configured delay does not establish recovery. Unconfirmed recovery is reported as right-censored within the study horizon.

• Missing current/head/level readings remain indeterminate where unsupported or unavailable. Old inspection values retain their timestamp; they cannot justify the current snapshot until explicitly refreshed.

• Counterfactual proofs include effective station inputs, the exact changed override patch, solver status and flow/power changes. Other faults remain in place; a masked repair can yield no flow improvement.

• Numerical observation/calculation tolerances are explicit demonstration rules, not calibrated statistical limits. Quality indices are not probabilities. Vibration/temperature references are labeled illustrative formulas, not validated health predictions.

• Permanently lost production, calibrated failure probabilities, motor amperes, fault/arc-flash/protection analysis, specialist 3D flood routing and uncalibrated reliability predictions remain explicitly unsupported.

Verification

src/tests/engineeringIntelligence.test.ts checks the runtime inventory, complete analysis proofs, unknown-field detection, unsupported requests and recorded historical assumptions. e2e/intelligence.spec.ts opens diagnostic, recovery, challenge, plant and forecast proof panels against real UI results. The full test counts and current execution results are recorded in ENGINEERING-VERIFICATION.md.

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